Linear Regression and Feature Selection

Assoc. Prof. Quan Thanh Tho

Learning Objectives

Regression - estimation of the relationship between variables

- Linear regression
- Assessing the assumptions
- Non-linear regression

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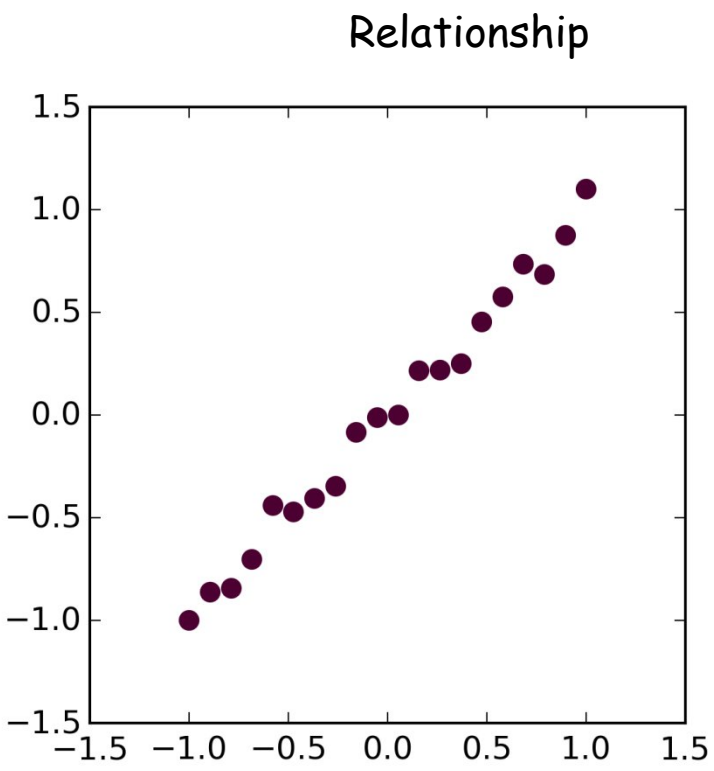
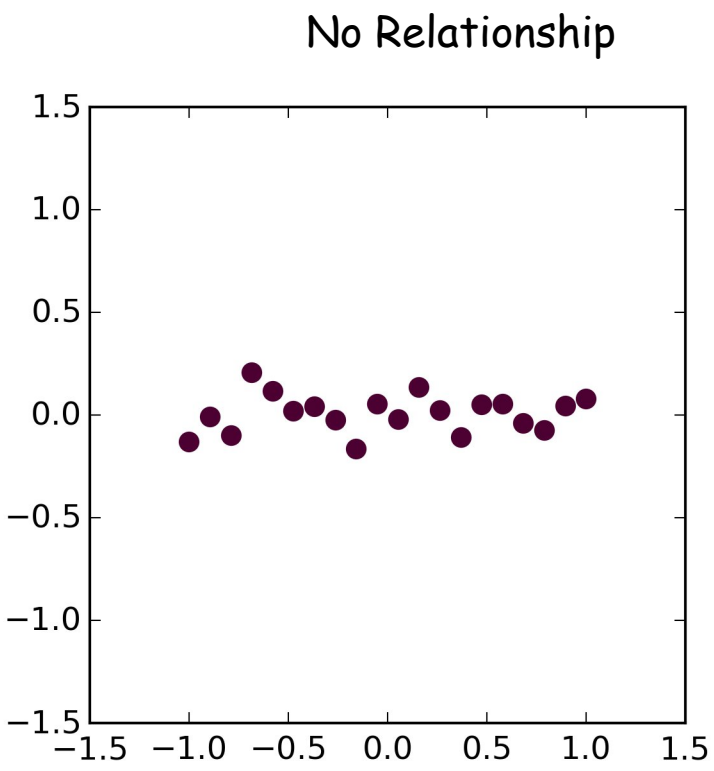
Correlation

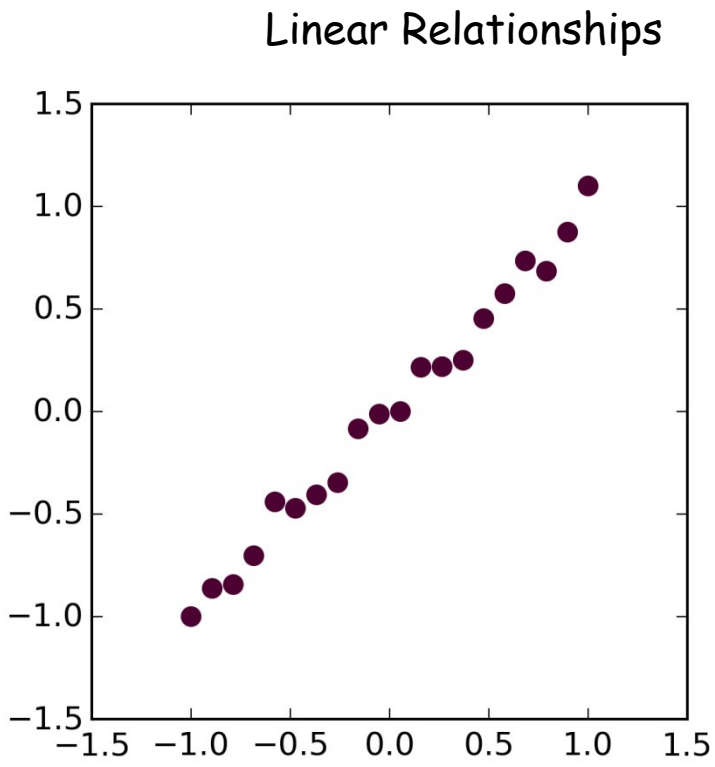
- Correlation coefficient quantifies the association strength
- Sensitivity to the distribution

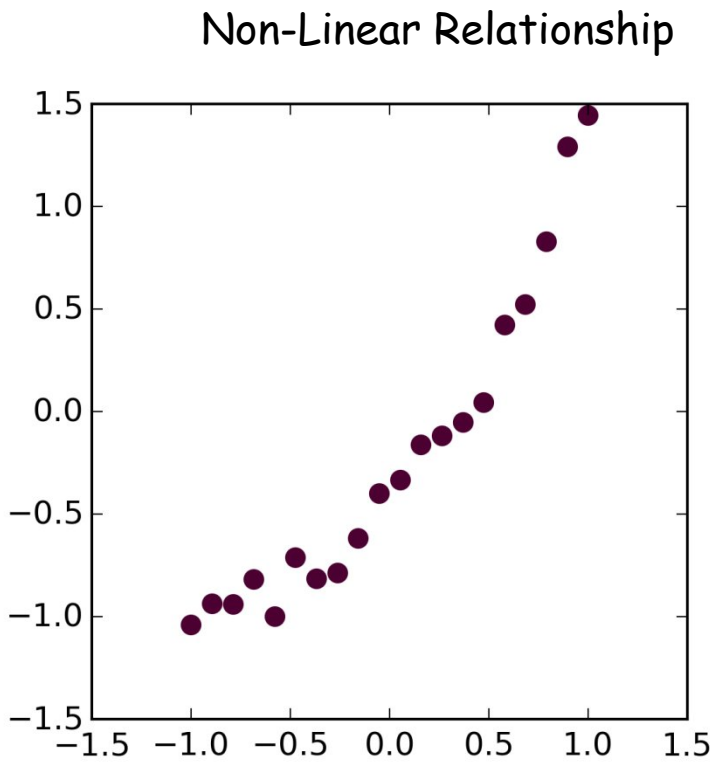
Relationships





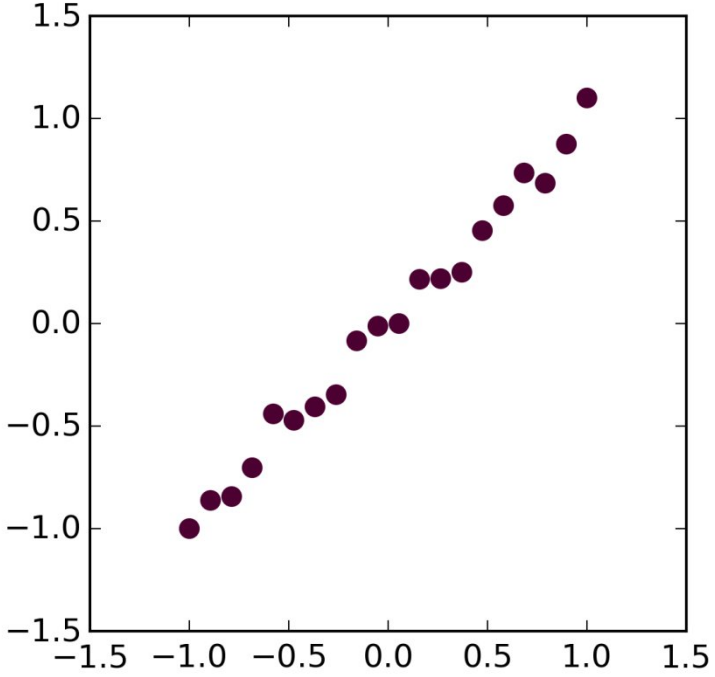
Relationships



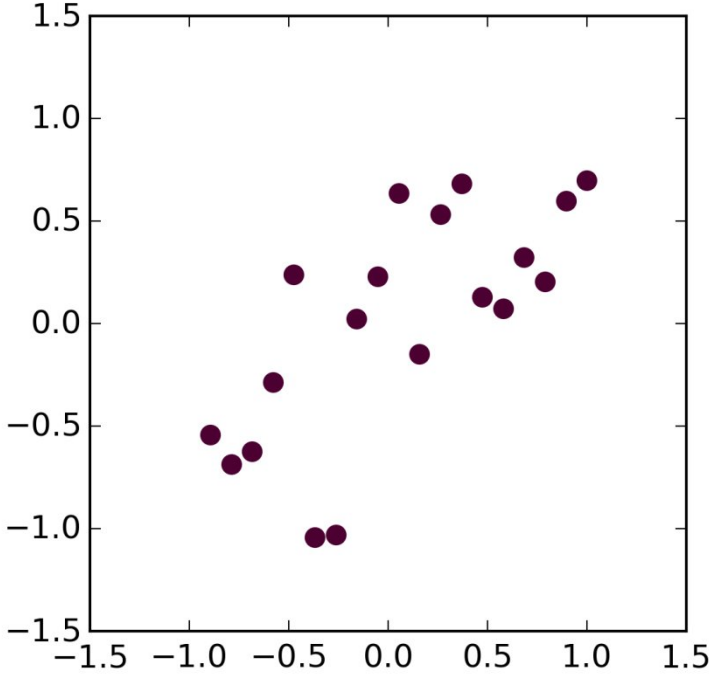


Relationships

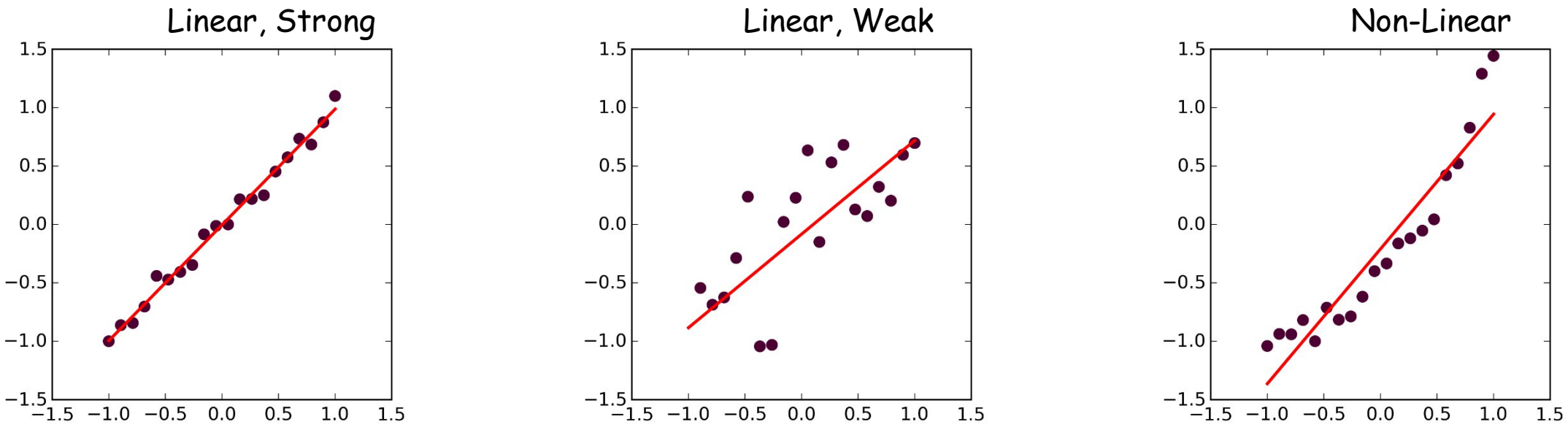
Linear, Strong



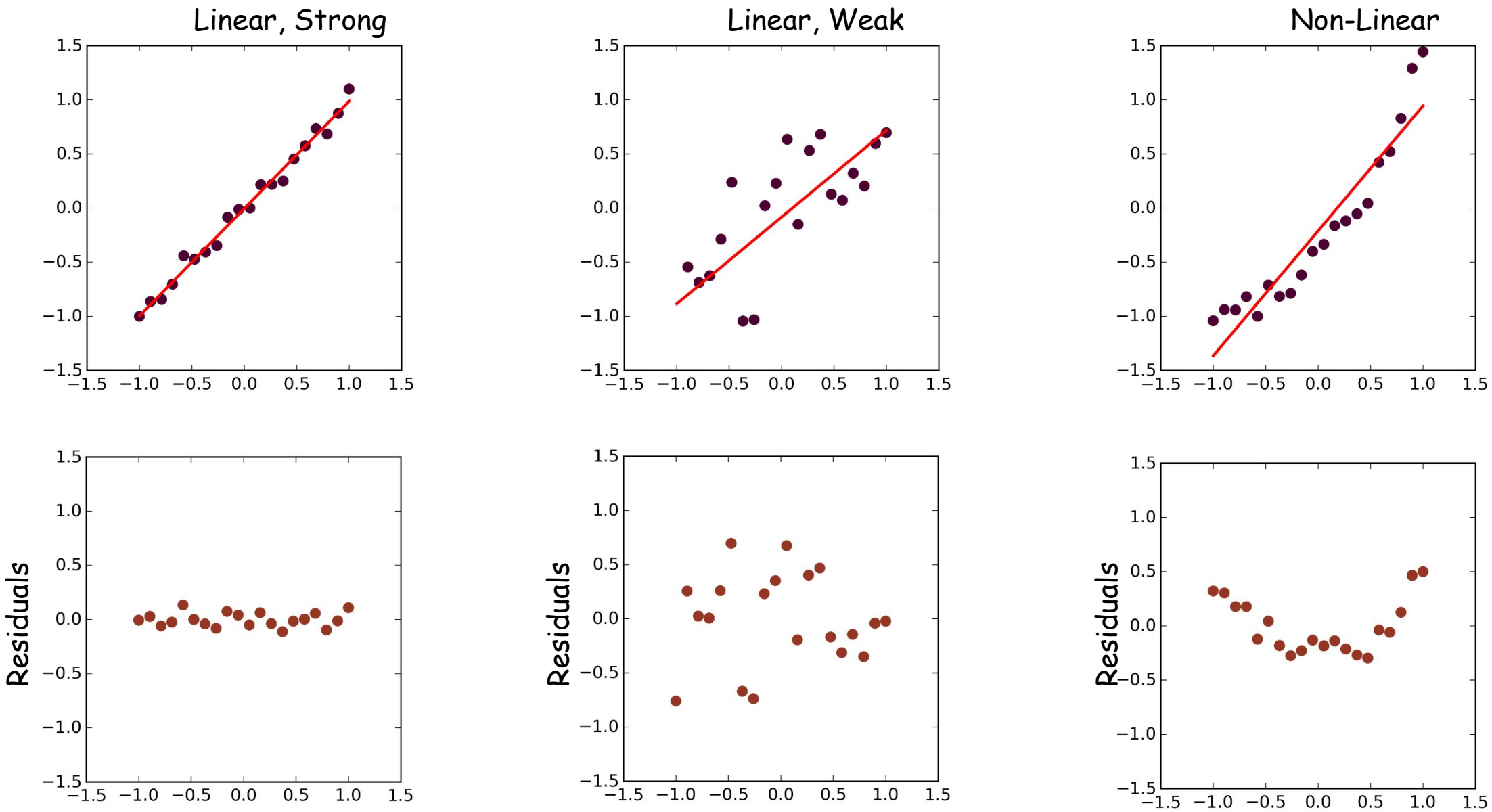
Linear, Weak



Linear Regression



Linear Regression - Residuals



Linear Regression Model

Intercept

Slope

Independent Variable

Dependent Variable

$$\underbrace{Y_i = \beta_0 + \beta_1 X_i}_{\text{Linear component}} + \underbrace{\varepsilon_i}_{\text{Random Error component}}$$

Random Error

Linear Regression Assumptions

- The relationship between the variables is linear.

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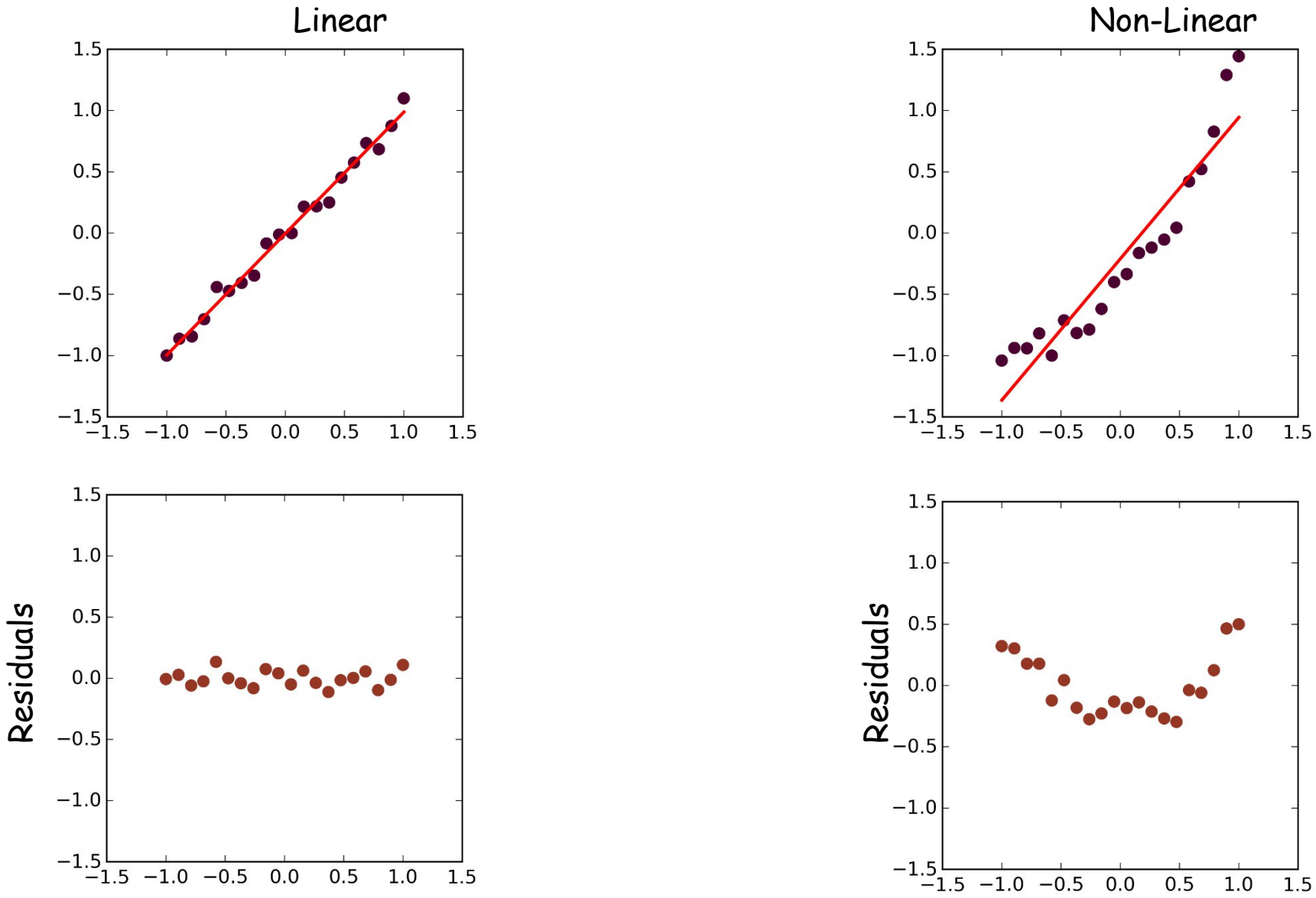
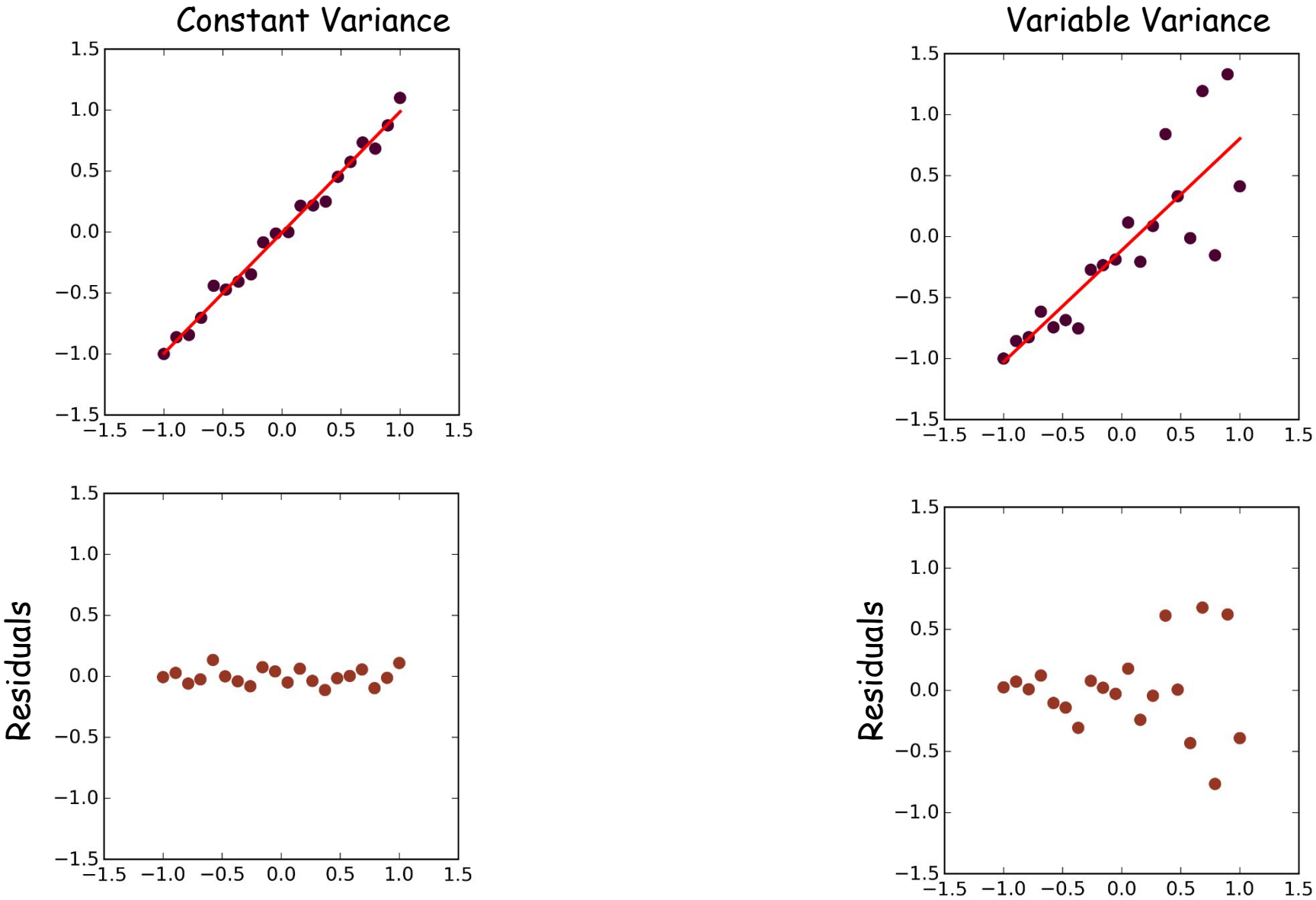
- The relationship between the variables is linear.
- Errors are independent, normally distributed with mean zero and constant variance.

Linear Regression Assumptions



Linear Regression Assumptions



Linear Regression Model

Intercept

Slope

Independent Variable

Dependent Variable

$$\underbrace{Y_i = \beta_0 + \beta_1 X_i}_{\text{Linear component}} + \underbrace{\varepsilon_i}_{\text{Random Error component}}$$

Random Error

Linear Regression - Estimating the line

Estimated Intercept

Estimated Slope

Estimated Value

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

Independent Variable

Least Squares Method

Find slope and intercept given measurements X_i, Y_i, $i=1..N$ that minimizes the sum of the squares of the residuals.

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$$S = \sum \varepsilon_i^2$$

Least Squares Method

Find slope and intercept given measurements X_i, Y_i, $i=1..N$ that **minimizes** the sum of the squares of the residuals.

$$\frac{\partial S}{\partial \hat{\beta}_0} = 0 \qquad \frac{\partial S}{\partial \hat{\beta}_1} = 0$$

Least Squares Method

Find slope and intercept given measurements X_i, Y_i, $i=1..N$ that **minimizes the sum of the squares of the residuals.**

$$\frac{\partial S}{\partial \hat{\beta}_1} = \frac{\partial}{\partial \hat{\beta}_1} \sum \varepsilon_i^2 = \frac{\partial}{\partial \hat{\beta}_1} \sum (Y_i - \hat{Y}_i)^2 = \frac{\partial}{\partial \hat{\beta}_1} \sum (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2 = \frac{\partial}{\partial \hat{\beta}_1} \sum (Y_i^2 - 2Y_i(\hat{\beta}_0 + \hat{\beta}_1 X_i) + (\hat{\beta}_0 + \hat{\beta}_1 X_i)^2) =$$

$$= \frac{\partial}{\partial \hat{\beta}_1} \sum (Y_i^2 - 2Y_i(\hat{\beta}_0 + \hat{\beta}_1 X_i) + \hat{\beta}_0^2 + 2\hat{\beta}_0\hat{\beta}_1 X_i + \hat{\beta}_1^2 X_i^2) = \frac{\partial}{\partial \hat{\beta}_1} \sum (Y_i^2 - 2Y_i(\hat{\beta}_0 + \hat{\beta}_1 X_i) + \hat{\beta}_0^2 + 2\hat{\beta}_0\hat{\beta}_1 X_i + \hat{\beta}_1^2 X_i^2) =$$

$$= \frac{\partial}{\partial \hat{\beta}_1} \sum (-2Y_i\hat{\beta}_1 X_i + 2\hat{\beta}_0\hat{\beta}_1 X_i + \hat{\beta}_1^2 X_i^2) = \sum (-2Y_i X_i + 2\hat{\beta}_0 X_i + 2\hat{\beta}_1 X_i^2) =$$

$$= \sum (-2Y_i X_i + 2(\overline{Y} - \hat{\beta}_1 \overline{X}) X_i + 2\hat{\beta}_1 X_i^2) = -2\sum Y_i X_i + 2\overline{Y}\sum X_i - 2\hat{\beta}_1 \overline{X}\sum X_i + 2\hat{\beta}_1 \sum X_i^2 =$$

$$= -2\sum Y_i X_i + 2\frac{\sum Y_i}{N}\sum X_i - 2\hat{\beta}_1 \frac{\sum X_i}{N}\sum X_i + 2\hat{\beta}_1 \sum X_i^2 = 2(-\sum Y_i X_i + \frac{\sum Y_i \sum X_i}{N} - \hat{\beta}_1 (\frac{(\sum X_i)^2}{N} - \sum X_i^2)) = 0$$

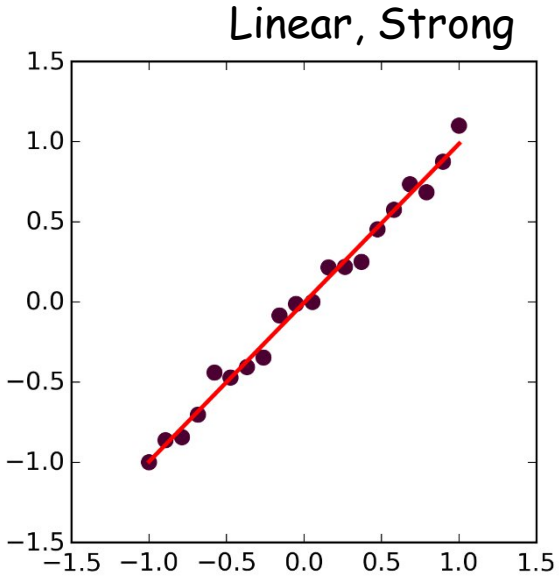
$$\hat{\beta}_1 = \frac{\frac{\sum Y_i \sum X_i}{N} - \sum Y_i X_i}{\frac{(\sum X_i)^2}{N} - \sum X_i^2}$$

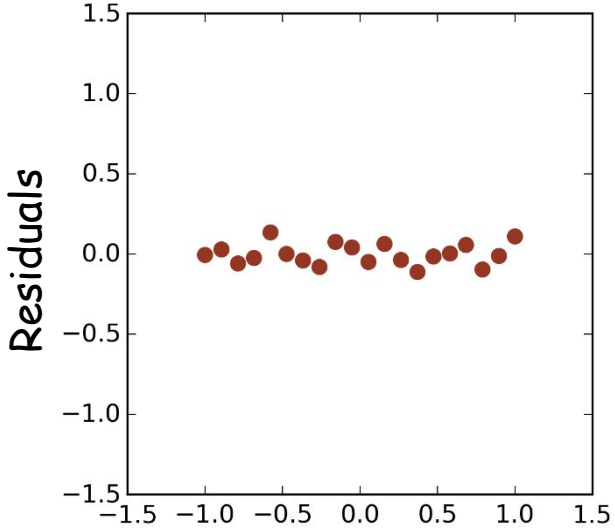
$$\hat{\beta}_0 = \frac{\sum Y_i}{N} - \hat{\beta}_1 \frac{\sum X_i}{N}$$

Linear Regression in Python

```
import scipy.stats as stats

slope,intercept,r_value,p_value,std_err = stats.linregress(x,y)
```

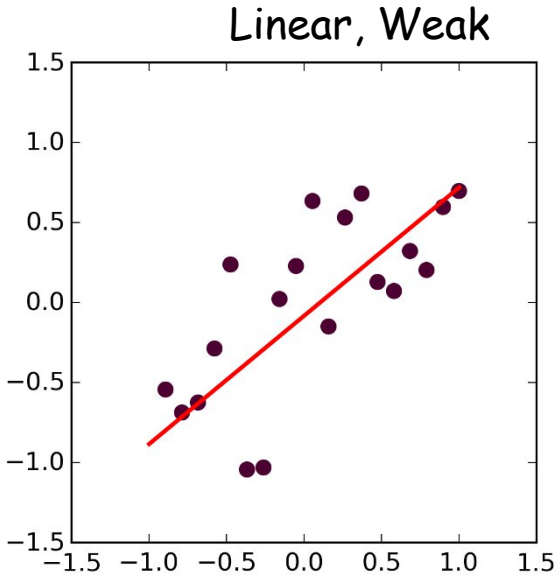
Linear Regression Example

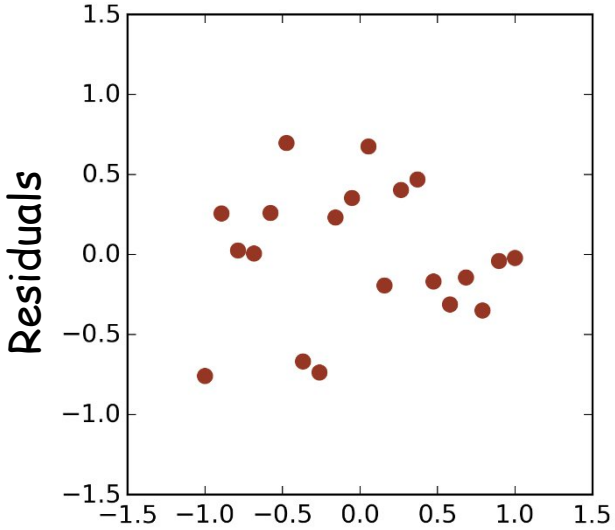





```
x=np.linspace(-1,1,points)
y=x+0.1*np.random.normal(size=points)
slope,intercept,r_value,p_value,std_err   =
stats.linregress(x,y)
y_line=slope*x+intercept

fig, (ax1) = plt.subplots(1,figsize=(4,4))
ax1.scatter(x,y,color='#4D0132',lw=0,s=60)
ax1.set_xlim([-1.5,1.5])
ax1.set_ylim([-1.5,1.5])
ax1.plot(x,y_line,color='red',lw=2)
fig.savefig('linear.png')

fig, (ax1) = plt.subplots(1,figsize=(4,4))
ax1.scatter(x,y-y_line, color='#963725',lw=0,s=60)
ax1.set_xlim([-1.5,1.5])
ax1.set_ylim([-1.5,1.5])
fig.savefig('linear-residuals.png')
```

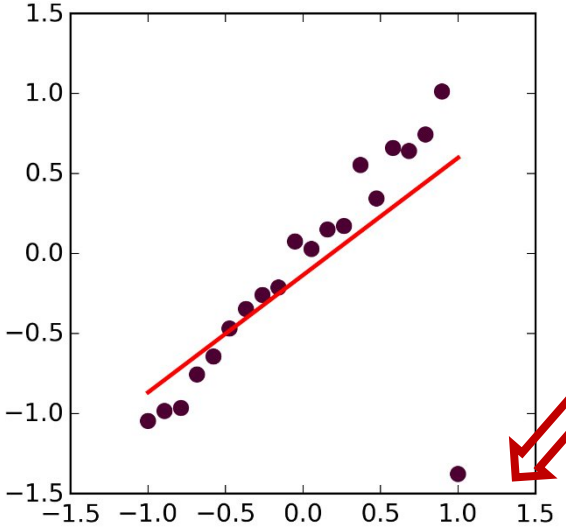
Linear Regression Example



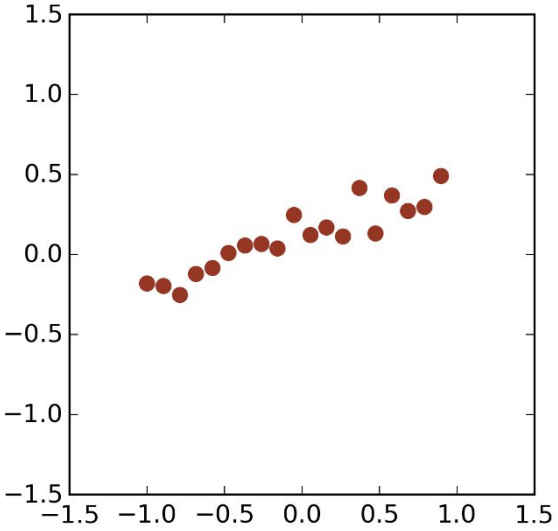



```
x=np.linspace(-1,1,points)
y=x+0.4*np.random.normal(size=points)
slope,intercept,r_value,p_value,std_err   =
stats.linregress(x,y)
y_line=slope*x+intercept

fig, (ax1) = plt.subplots(1,figsize=(4,4))
ax1.scatter(x,y,color='#4D0132',lw=0,s=60)
ax1.set_xlim([-1.5,1.5])
ax1.set_ylim([-1.5,1.5])
ax1.plot(x,y_line,color='red',lw=2)
fig.savefig('linear-weak.png')

fig, (ax1) = plt.subplots(1,figsize=(4,4))
ax1.scatter(x,y-y_line, color='#963725',lw=0,s=60)
ax1.set_xlim([-1.5,1.5])
ax1.set_ylim([-1.5,1.5])
fig.savefig('linear-weak-residuals.png')
```

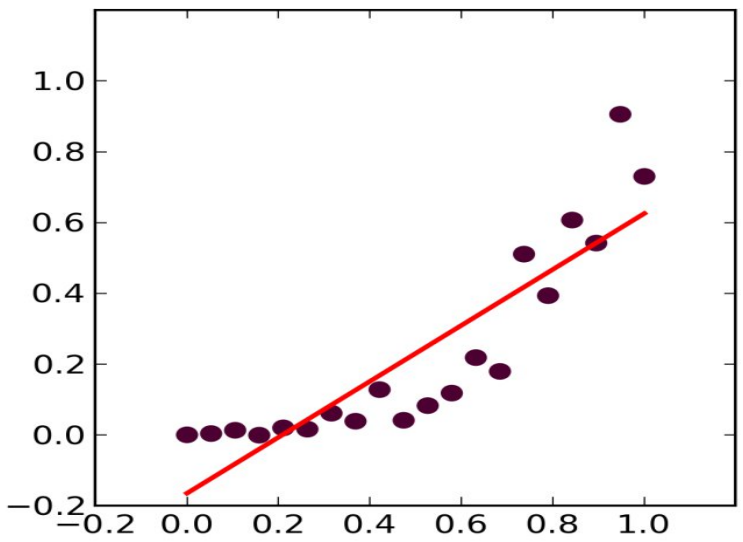
Linear Regression Example



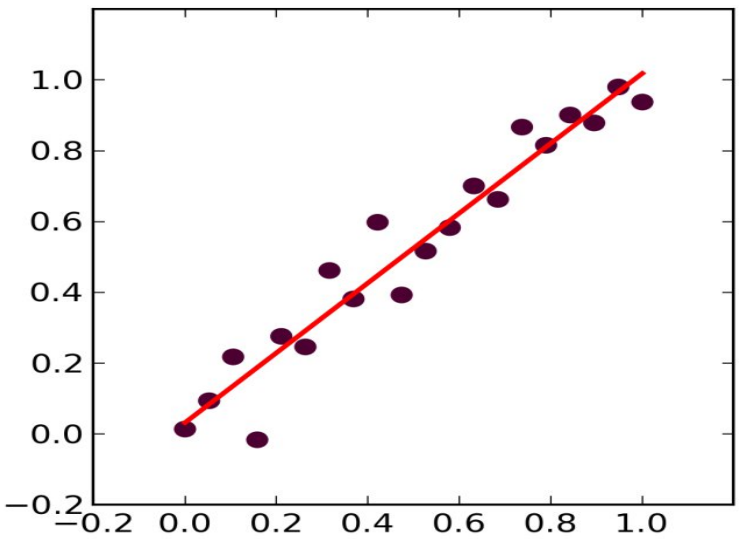
Outlier



Regression - Non-linear data



Solution 1: Transformation



Solution 2: Non-linear Regression

$$\boxed{\hat{Y}_i = f(X_i, \hat{\beta}_0, \hat{\beta}_1, \ldots)}$$

Regularization

- Maximum-likelihood estimates are not always the best
- Alternative: we "regularize" the likelihood objective (also known as penalized likelihood, shrinkage, smoothing, etc.), by adding to it a penalty term:

$$\hat{\theta}_{\text{shrinkage}} = \arg\min_{\theta}\left[-l(\theta;D) + \lambda\|\theta\|\right]$$

  - where ||θ|| might be the L1 or L2 norm.

- The choice of norm has an effect
  - using the L2 norm pulls directly towards the origin,
  - while using the L1 norm pulls towards the coordinate axes, i.e it tries to set some of the coordinates to 0.
  - This second approach can be useful in a feature-selection setting.

Regularized Regression

- Recall linear regression:

$$\mathbf{y} = \mathbf{X}^T \beta + \varepsilon$$

$$\beta^* = \arg\max_\beta (\mathbf{y} - \mathbf{X}^T \beta)^T (\mathbf{y} - \mathbf{X}^T \beta)$$

$$= \arg\max_\beta \| \mathbf{y} - \mathbf{X}^T \beta \|^2$$

- Regularized LR:
  - L2-regularized LR:

$$\beta^* = \arg\max_\beta \| \mathbf{y} - \mathbf{X}^T \beta \|^2 + \lambda \| \beta |$$

where

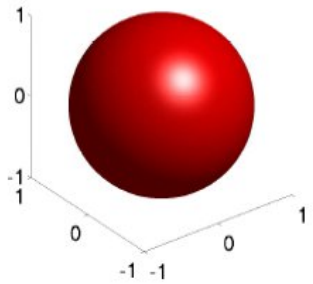
$$\| \beta \| = \sum_i \beta_i^2$$

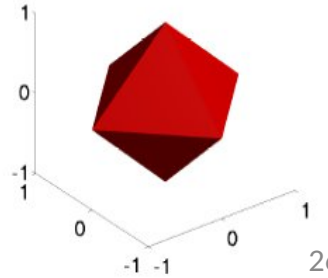
  - L1-regularized LR:

$$\beta^* = \arg\max_\beta \| \mathbf{y} - \mathbf{X}^T \beta \|^2 + \lambda | \beta |$$

where

$$| \beta | = \sum_i | \beta_i |$$





Performs a model selection directly

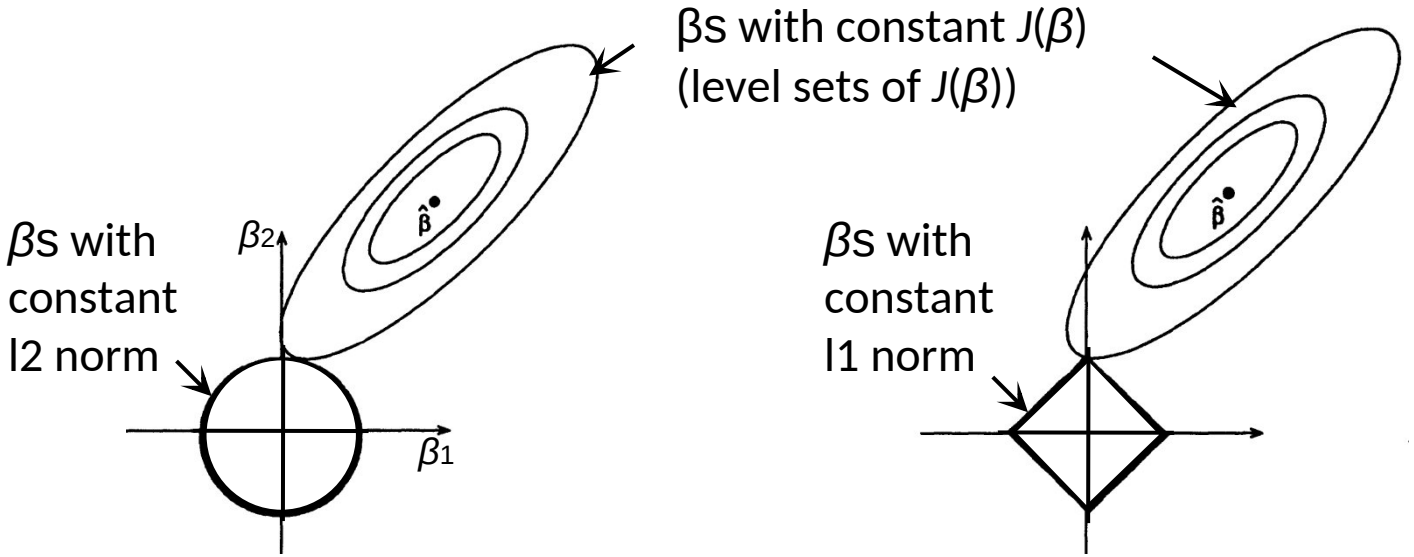
Ridge Regression vs Lasso

$$\min_{\beta}(\mathbf{X}\beta - \mathbf{Y})^T(\mathbf{X}\beta - \mathbf{Y}) + \lambda \text{pen}(\beta) = \min_{\beta} J(\beta) + \lambda \text{pen}(\beta)$$

Ridge Regression:

$$\text{pen}(\beta) = \|\beta\|_2^2$$

Lasso:

$$\text{pen}(\beta) = \|\beta\|_1$$

HOT !